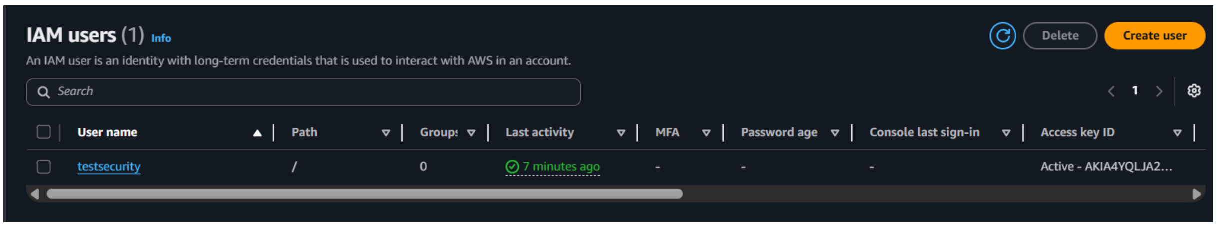
AWS Configuration Guide

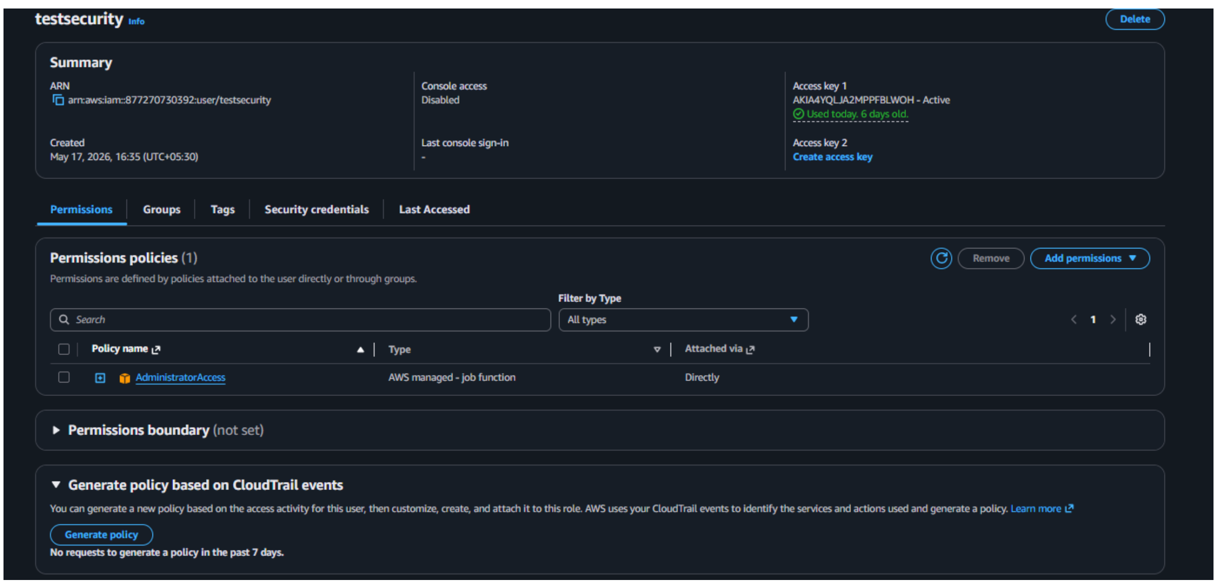
Network Security ML Project — Cloud Setup for S3, ECR & EC2 Deployment

1. IAM — Identity & Access Management

An IAM user was created with programmatic access to allow secure, key-based interaction with AWS services (S3, ECR, EC2) from the local environment and CI/CD pipeline, instead of using root account credentials.



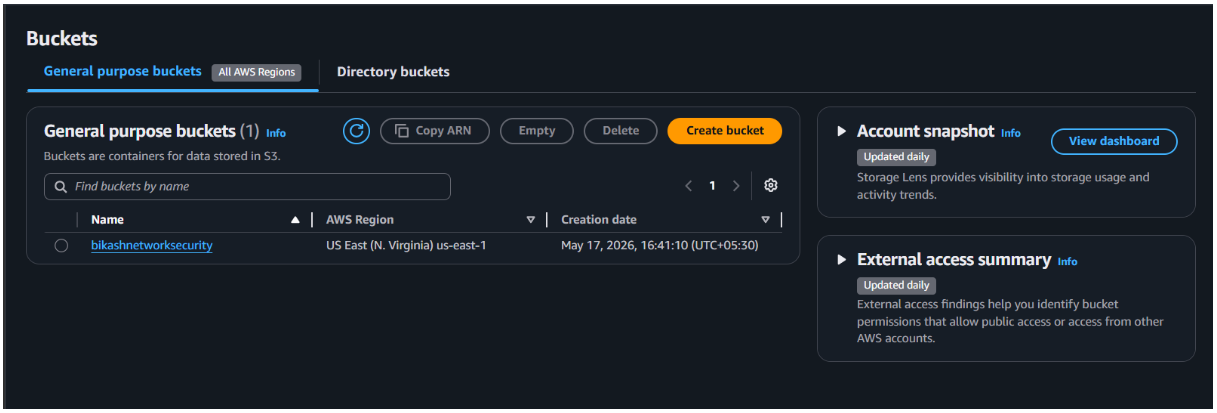
IAM user overview



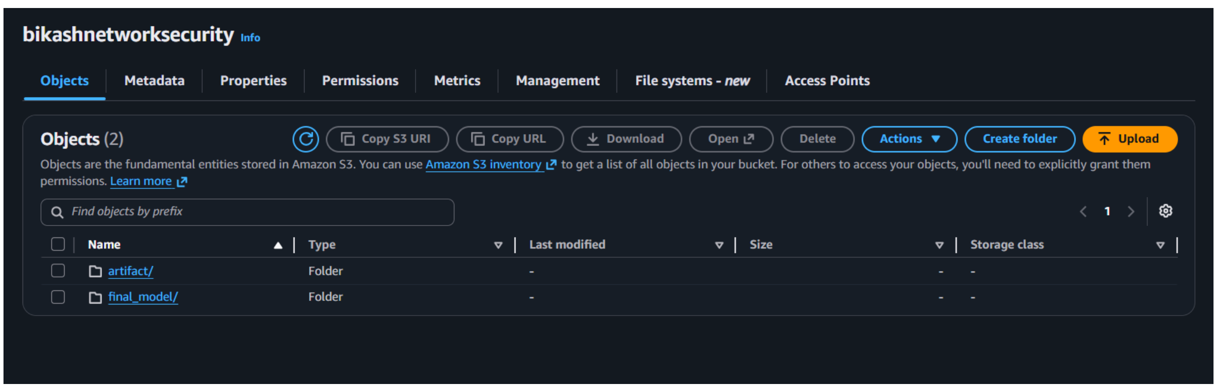
IAM user permissions and access key configuration

2. S3 — Simple Storage Service

An S3 bucket was configured to store pipeline artifacts and the final trained model, enabling the training pipeline to sync both the intermediate artifact directory and the production-ready model to the cloud after every run.



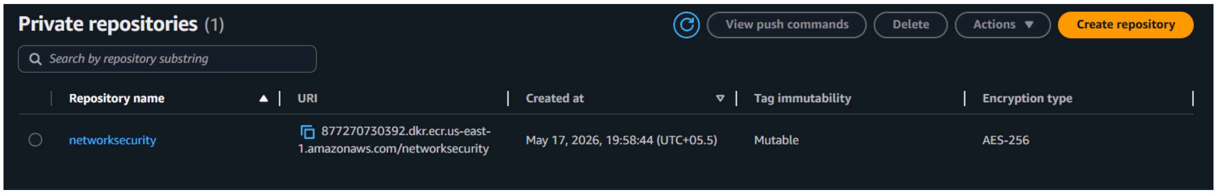
S3 bucket configuration



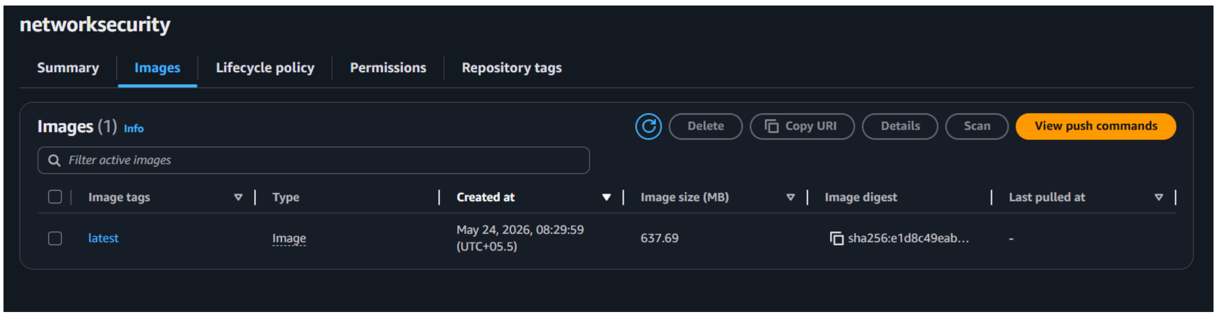
S3 bucket contents — synced artifacts and model files

3. ECR — Elastic Container Registry

A private ECR repository was created to store the Docker image of the application. GitHub Actions builds this image on every push and pushes it here, making it available for deployment to EC2.



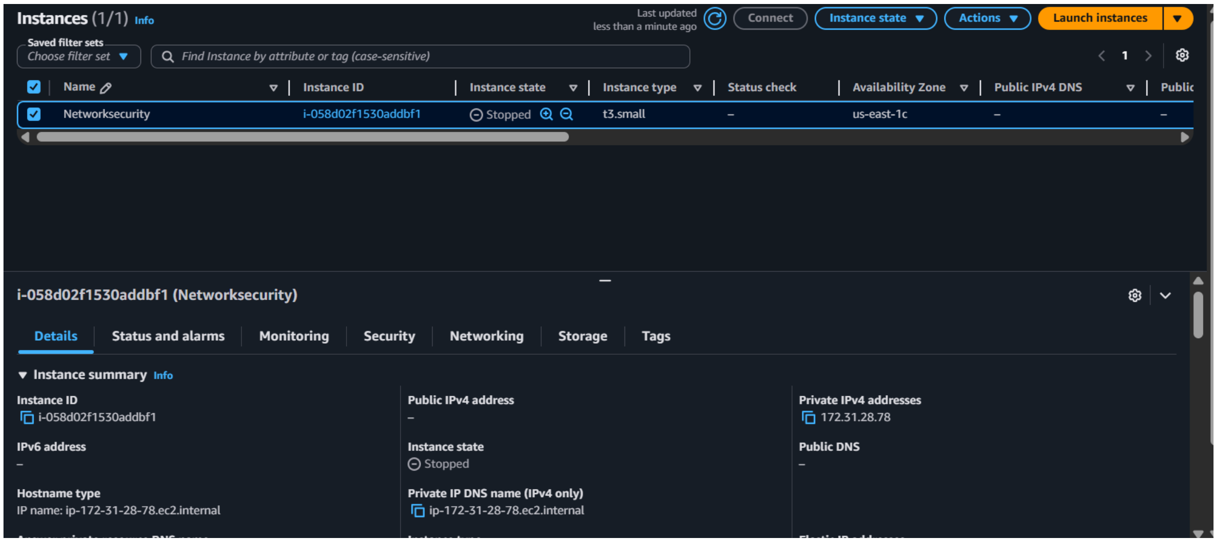
ECR repository overview



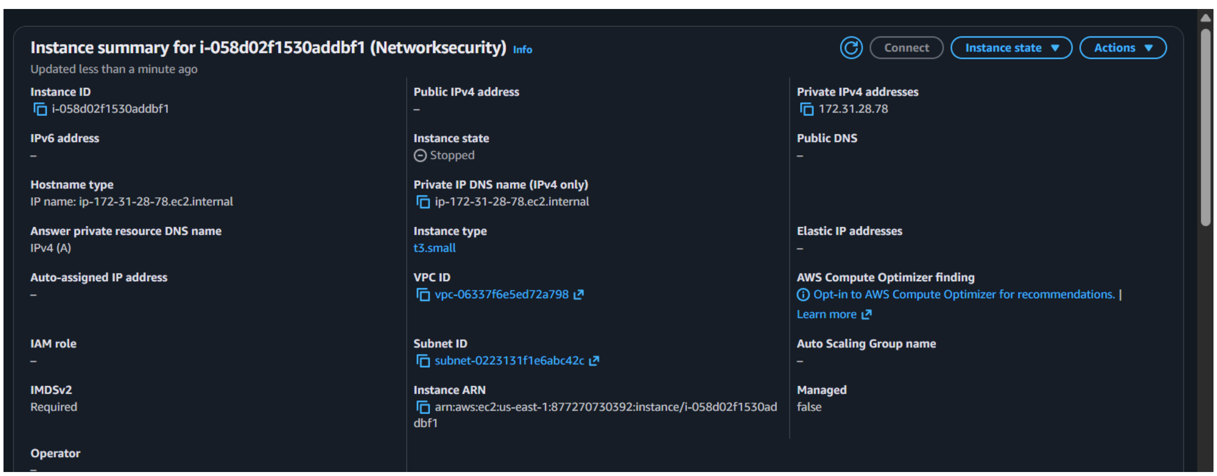
ECR — pushed Docker image with tag details

4. EC2 — Elastic Compute Cloud

An EC2 instance (Ubuntu) was launched to host the live application. Docker was installed on the instance, and a self-hosted GitHub Actions runner was configured so the CI/CD pipeline could pull the latest image from ECR and redeploy the container automatically.



EC2 instance details



EC2 — security group / running configuration

Pipeline: IAM (access) → S3 (storage) → ECR (image registry) → EC2 (live deployment)